DBMS – SQL QUERY IMPLEMENTATION

Project Title: Duke Bike Inventory and Customer Order Management System

Objectives

1. Perform SELECT queries to retrieve product and customer information.
2. Filter product records using WHERE conditions.
3. Apply AND, OR, IN, and BETWEEN operators.
4. Sort product records using ORDER BY.
5. Generate customer-wise order count and total purchase reports.

SQL Queries and Output Screenshots

The following sections contain the SQL commands and output screenshots provided.

1. SELECT QUERY – Display All Products

Displays all records from the Product table, including product ID, product name, category, price, and stock.

```
SQL> SELECT * FROM Product;

 PRODUCTID PRODUCTNAME
---------- --------------------------------------------------
CATEGORY                         PRICE      STOCK
------------------------------ ---------- ----------
       101 Duke 200
Bike                             200000          5

       102 Duke 250
Bike                             230000          3

       103 Duke 390
Bike                             320000          2


 PRODUCTID PRODUCTNAME
---------- --------------------------------------------------
CATEGORY                         PRICE      STOCK
------------------------------ ---------- ----------
       104 Helmet
Accessories                        2500         10

       105 Bike Gloves
Accessories                        1500          0

       106 Engine Oil
Spare Parts                         800         15


6 rows selected.
```

2. SELECT QUERY – Display Selected Product Columns

Retrieves ProductID, ProductName, Price, and Stock from the Product table.

```
SQL> SELECT ProductID, ProductName, Price, Stock
  2  FROM Product;

 PRODUCTID PRODUCTNAME                                          PRICE
---------- -------------------------------------------------- ----------
     STOCK
----------
       101 Duke 200                                           200000
         5

       102 Duke 250                                           230000
         3

       103 Duke 390                                           320000
         2


 PRODUCTID PRODUCTNAME                                          PRICE
---------- -------------------------------------------------- ----------
     STOCK
----------
       104 Helmet                                               2500
        10

       105 Bike Gloves                                          1500
         0

       106 Engine Oil                                            800
        15


6 rows selected.
```

3. WHERE Clause – Filter by Price

Shows the output for products priced below ₹1000 and above ₹1000.

```
SQL> SELECT * FROM Product
  2  WHERE Price < 1000;

 PRODUCTID PRODUCTNAME
---------- --------------------------------------------------
CATEGORY                         PRICE      STOCK
------------------------------ ---------- ----------
       106 Engine Oil
Spare Parts                        800         15


SQL> SELECT * FROM Product
  2  WHERE Price > 1000;

 PRODUCTID PRODUCTNAME
---------- --------------------------------------------------
CATEGORY                         PRICE      STOCK
------------------------------ ---------- ----------
       101 Duke 200
Bike                             200000          5

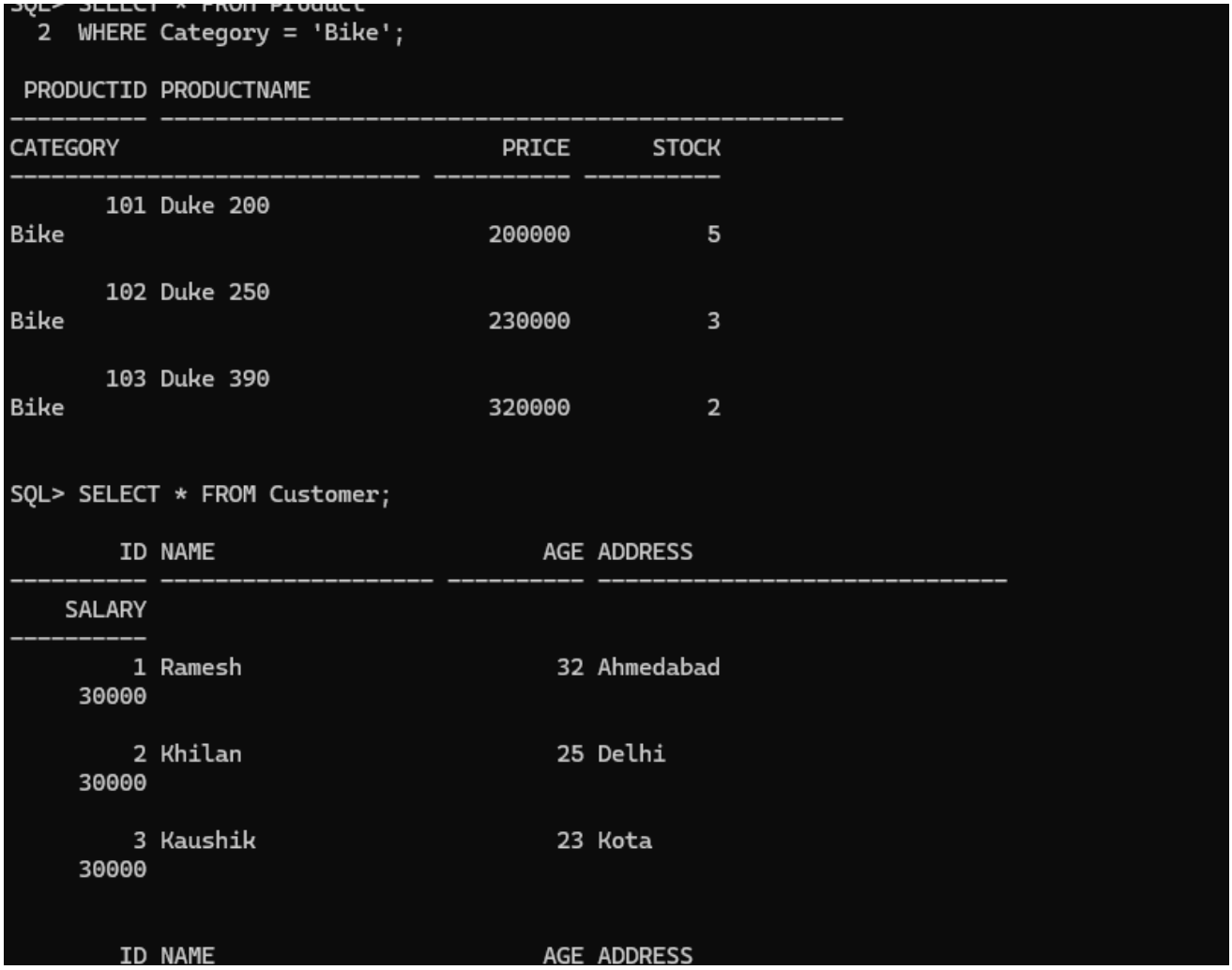
       102 Duke 250
Bike                             230000          3

       103 Duke 390
Bike                             320000          2


 PRODUCTID PRODUCTNAME
---------- --------------------------------------------------
CATEGORY                         PRICE      STOCK
------------------------------ ---------- ----------
       104 Helmet
Accessories                        2500         10
```

4. WHERE Clause – Filter by Category and Display Customers

Shows products in the Bike category and the Customer table output.



```
SQL> SELECT * FROM Product
  2  WHERE Category = 'Bike';

 PRODUCTID PRODUCTNAME
---------- --------------------------------------------------
CATEGORY                         PRICE      STOCK
------------------------------ ---------- ----------
       101 Duke 200
Bike                           200000          5

       102 Duke 250
Bike                           230000          3

       103 Duke 390
Bike                           320000          2


SQL> SELECT * FROM Customer;

        ID NAME                      AGE ADDRESS
---------- -------------------- ---------- ------------------------------
    SALARY
----------
         1 Ramesh                        32 Ahmedabad
     30000

         2 Khilan                        25 Delhi
     30000

         3 Kaushik                       23 Kota
     30000


        ID NAME                      AGE ADDRESS
```

5. Customer Search and AND / OR Conditions

Shows a search for customer Ramesh, a combined price-and-stock filter, and a category filter using OR.

```
SQL> SELECT * FROM Customer
  2  WHERE NAME = 'Ramesh';

        ID NAME                           AGE ADDRESS
---------- -------------------- ---------- --------------------
    SALARY
----------
         1 Ramesh                         32 Ahmedabad
     30000


SQL> SELECT * FROM Product
  2  WHERE Price < 3500 AND Stock > 5;

 PRODUCTID PRODUCTNAME
---------- --------------------------------------------------
CATEGORY                          PRICE      STOCK
------------------------------ ---------- ----------
       104 Helmet
Accessories                         2500         10

       106 Engine Oil
Spare Parts                          800         15


SQL> SELECT * FROM Product
  2  WHERE Category = 'Bike'
  3     OR Category = 'Accessories';

 PRODUCTID PRODUCTNAME
```

6. IN Operator and BETWEEN Operator

Shows product filtering by category using IN and price range using BETWEEN.

```
 PRODUCTID PRODUCTNAME
---------- --------------------------------------------------
CATEGORY                         PRICE      STOCK
------------------------------ ---------- ----------
       101 Duke 200
Bike                            200000          5

       102 Duke 250
Bike                            230000          3

       103 Duke 390
Bike                            320000          2


 PRODUCTID PRODUCTNAME
---------- --------------------------------------------------
CATEGORY                         PRICE      STOCK
------------------------------ ---------- ----------
       104 Helmet
Accessories                       2500         10

       105 Bike Gloves
Accessories                       1500          0


SQL> SELECT * FROM Product
  2  WHERE Category IN ('Bike', 'Accessories');

 PRODUCTID PRODUCTNAME
---------- --------------------------------------------------
CATEGORY                         PRICE      STOCK
------------------------------ ---------- ----------
```

7. ORDER BY – Sort Products by Price

Shows the beginning of the query that sorts products by price in descending order.

```
 PRODUCTID PRODUCTNAME
---------- --------------------------------------------------
CATEGORY                         PRICE      STOCK
------------------------------ ---------- ----------
       104 Helmet
Accessories                       2500         10

       105 Bike Gloves
Accessories                       1500          0


SQL> SELECT * FROM Product
  2  WHERE Price BETWEEN 1000 AND 10000;

 PRODUCTID PRODUCTNAME
---------- --------------------------------------------------
CATEGORY                         PRICE      STOCK
------------------------------ ---------- ----------
       104 Helmet
Accessories                       2500         10

       105 Bike Gloves
Accessories                       1500          0


SQL> SELECT ProductID, ProductName, Price
  2  FROM Product
  3  ORDER BY Price DESC;

 PRODUCTID PRODUCTNAME                                         PRICE
```

8. Customer-wise Order and Purchase Reports

Shows customer-wise total order counts and total purchase amounts using LEFT JOIN and GROUP BY.

```
SQL> SELECT c.ID, c.NAME,
  2         COUNT(o.Order_ID) AS Total_Orders
  3  FROM Customer c
  4  LEFT JOIN Orders o
  5  ON c.ID = o.Customer_ID
  6  GROUP BY c.ID, c.NAME;

        ID NAME                 TOTAL_ORDERS
---------- -------------------- ------------
         1 Ramesh                          0
         2 Khilan                          0
         4 Chaitali                        0
         5 Hardik                          0
         3 Kaushik                         0

SQL> SELECT c.ID, c.NAME,
  2         NVL(SUM(o.Total_Amount), 0) AS Total_Purchase
  3  FROM Customer c
  4  LEFT JOIN Orders o
  5  ON c.ID = o.Customer_ID
  6  GROUP BY c.ID, c.NAME;

        ID NAME                 TOTAL_PURCHASE
---------- -------------------- --------------
         1 Ramesh                            0
         2 Khilan                            0
         4 Chaitali                          0
         5 Hardik                            0
         3 Kaushik                           0
```